Three Nulls and the Shape They Make

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Cross-arc joint synthesis (second) **Date:** 2026-07-27 **Scope:** DCR3 + DCR3b + Constraint-Swap intervention-algebra reanalysis. All three preregistered. All three NO_GO. They fail in the same direction, at the same structural feature. That's the finding.

The three nulls

deletion) ranks 3rd at 1904, behind unclassified and T2. NO_GO.

dependence *within the corpus*. T1 still ranks 3rd (score 20, T2 at 39, unclassified at 725). NO_GO.

R3 NO_GO. Correlations low (r_undo = −0.234, r_rescue = 0.079) but permutation p ≈ 0.10, not < 0.05. Marginal.

- **DCR3** (PR #442): DR-arc corpus-frequency nominator. T1 (Einstein's
- **DCR3b**: replaced frequency with LLM-scored counterfactual
- **Constraint-Swap intervention-algebra reanalysis**: R1 GO, R2 GO,

The shape they make

The human director's reframe proposed the target commitment lives in the **intervention algebra** — the family of interventions under which behavior changes — not inside the visible representation. Three tests. All three failed at their strictest preregistered gate.

Correction (2026-07-27, after human-director critique). An earlier version of this section wrote "*the object is outside the visible representation the measurement can access*" and treated the three failures as instances of the *same* structural feature. That was too strong on two counts and I'm correcting both:

physicists didn't write down what they globally presupposed — a linguistic/pragmatic phenomenon. Constraint Swap fails because a specific hypothesized geometric structure didn't manifest in that specific agent class under that specific design — an architectural/statistical phenomenon. Collapsing them into "the same failure" was cleaner as narrative than as evidence supports.

1. **Different failure mechanisms.** DCR fails because 1900

The tests establish that the measurement operators failed to recover the target. They do NOT distinguish among:

2. **Underdetermined between (B)/(C)/(D), not established as (A).**

- **(A)** The object literally lies outside the representation. - **(B)** The representation exists, but these operators cannot identify it. - **(C)** The object is distributed across representations. - **(D)** The target ontology is wrong.

For DCR, T1's realisations ARE in the DCR1e consensus (Lodge's "*time of journey… definite and independent of the motion,*" Larmor's common_time_across_two_systems, etc.) — the operators rank them low because they compete against T2 realisations that are more numerous and more explicitly cited. That's (B)/(C), not (A). For Constraint Swap, "geometry can't contain the object" overstates; the honest claim is "this particular geometric hypothesis was not supported in this agent class under this design" — again (B), possibly (D).

The narrower, defensible reading of the three nulls:

frequency and LLM within-corpus counterfactual scoring). That agreement IS a real pattern about the corpus: T2 is more explicitly cited than T1. It does not entail that T1 is irrecoverable from the corpus — only that these operators can't distinguish it from T2.

consistent with the reframe but underpowered at 32 seeds. Directionally supportive of intervention-algebra reasoning; not a positive confirmation.

- **DCR3 and DCR3b agree on rank order** (T2 > T1 under both
- **Constraint Swap reanalysis** shows correlations directionally

Why this is more informative than one null

Two independent scoring methods on the DCR corpus (frequency; LLM-judged counterfactual dependence) agree that T2 outweighs T1. They agree because they're both measuring quantities that live inside the corpus's explicit *citation* structure. T1 realisations exist in the corpus but are presupposed globally and cited locally by nothing.

For a scoring method to reach T1, it would need access to something neither DCR3 nor DCR3b provided: **inference about what each derivation IMPLICITLY REQUIRES, versus what it explicitly cites.** That's not a scoring-function upgrade. It's a different reasoning task, and it collides directly with DR7's soundness-completeness gap on open-realisation grouping functions.

The Constraint-Swap reanalysis produces a compatible signal in a different domain: the intervention effects, which the reframe predicted would be uncorrelated across seeds under constraint- specific structure, are *low-correlated* but not statistically distinguishable from a moderately-correlated null at 32 seeds. In both arcs, the reframe's prediction is directionally right but the measurement can't cross the preregistered threshold.

The refined honest claim

The intervention-algebra reframe is:

the observed data in all three tests

gate

actually decide it require capabilities the current instruments don't have (implicit-argument inference for DCR, larger seed count + intervention sweep for Constraint Swap)

- **not refuted** — its directional predictions are consistent with
- **not confirmed** — no test cleared its strictest preregistered
- **operationally constrained** — the measurements that would

That's an unusual result. Not "the idea is right" and not "the idea is wrong" — "the idea is testable in principle, and the specific tests we can currently run don't have the resolution to decide it." Rare shape. Worth naming as its own class of outcome.

What the pattern tells us that no single null could

Both arcs show the same failure mode: **corpus-statistic-based scoring for DCR, activation-geometry-based measurement for Constraint Swap** — the "object" always turns out to live outside what the measurement can see. But the reframe doesn't fully win either — you can't measure your way to it with the tools we currently have.

The remaining path forward:

derivation in the DCR corpus, LLM enumerates the premises the derivation *requires to be valid*, whether stated or not. Score each proposition by how often it appears in *inferred* premise sets. This is the semantic-access-to-D condition of DR6/DR6e operationalised on DCR.

with the same protocol, OR a random-transport sweep per seed with behavior-cluster analysis. Either would resolve the marginal R3 signal.

explicit premises has provably worse identification than one with access to inferred premises. Would generalise DCR3+DCR3b's shared failure mode into a corollary.

1. **DCR3c** — implicit-argument-inference scoring. For each
2. **Constraint Swap larger-N or intervention-sweep**. 100+ seeds
3. **DR9 theorem**: prove formally that any scoring restricted to

What survives, honestly

session (DCR3, DCR3b, DR6d, DR6e, Constraint Swap original, Constraint Swap reanalysis). Every one honored the gate verdict. Every one is reproducible byte-identically. Serial nulls under strict discipline are worth more than serial "reinterpretations."

predicted the shape of the null every time.

silent commitments, not loud ones. Extension-worthy to other cases with known ground truth.

- The methodology. Six load-bearing preregistered tests across the
- The theoretical framework (DR5 + DR7 + DR5+*DR5**). Correctly
- One substantive empirical claim from DCR3: revolutions delete

What doesn't

deletion. Not frequency, not LLM counterfactual dependence, not DR-arc kind-weight-times-degree.

as an untested direction, not a demonstrated one.

its mechanistic form — Constraint Swap already rejected it and the reanalysis's marginal signal doesn't rescue it.

- Any specific scoring function proposed to date reaches Einstein's
- The intervention-algebra reframe as a *positive* claim. It survives
- The "meaning is constraint-induced deformation" grand principle in

Where the ledger stands

Twenty-three papers merged this session. Six load-bearing preregistered nulls (with two GO papers — DCR1c/DCR1d — that established methodology and one GO on DCR2b that closed a scoping loop). The next serious empirical move on the DCR side is DCR3c (implicit-argument inference). The next serious move on the Constraint-Swap side is 100+ seed replication or a sweep-based reanalysis with raw hidden state data. Both are session-scoped in principle but neither has a preregistered plan yet.

The pattern is clean: preregister, run, report, repeat. Three serial nulls with the same failure mode tell us more about where the object is (outside the visible representation) than one clean GO would have told us about where it is (inside it).

That is the most a session can honestly produce.

Appendix: reproduction

Every experiment reproduces via its own runner. All prior arcs unchanged.